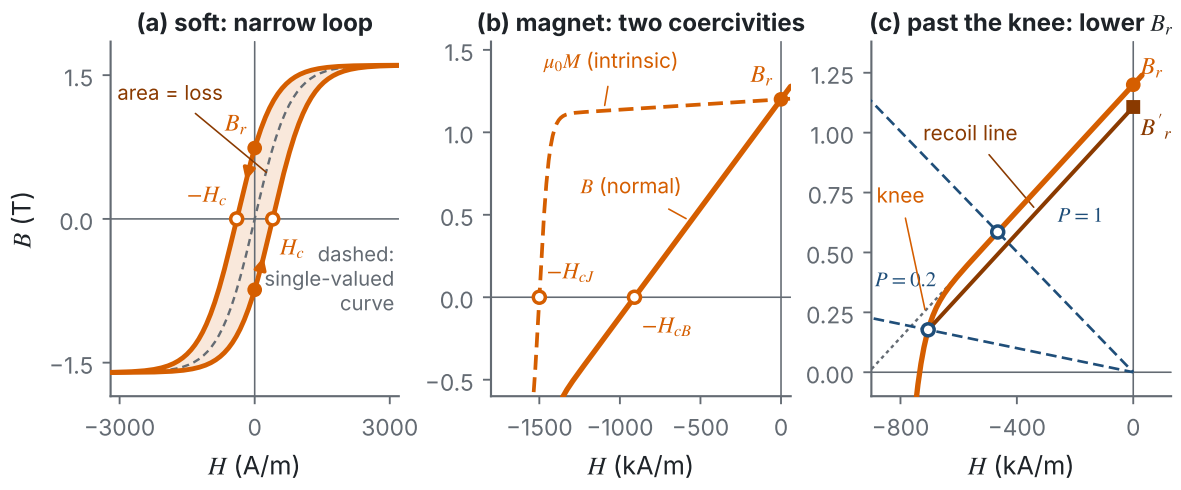

(a) soft: narrow loop
area = loss
B_r
$-H_c$
H_c
dashed: single-valued curve
B (T)
1.5
0.0
−1.5
−3000
0
3000
H (A/m)
(b) magnet: two coercivities
$\mu_0 M$ (intrinsic)
B_r
B (normal)
$-H_{cJ}$
$-H_{cB}$
1.5
1.0
0.5
0.0
−0.5
−1500
−1000
−500
0
H (kA/m)
(c) past the knee: lower B_r
B_r
B'_r
recoil line
knee
$P = 1$
$P = 0.2$
1.25
1.00
0.75
0.50
0.25
0.00
−800
−400
0
H (kA/m)